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Understanding secondary–tertiary transition in mathematics

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In Clark and Lovric (*Suggestion for a theoretical model for secondary–tertiary transition in mathematics*, Math. Educ. Res. J. 20(2) (2008), pp. 25–37) we began developing a model for the secondary–tertiary transition in mathematics, based on the anthropological notion of a rite of passage. We articulated several reasons why we believe that the educational transition from school to university mathematics should be viewed (and is) a rite of passage, and then examined certain aspects of the process of transition. Present article is a continuation of our study, resulting in an enhanced version of the model. In order to properly address all aspects of transition (such as a number of cognitive and pedagogical issues) we enrich our model with the notions of cognitive conflict (conceptual change) and culture shock (although defined and used in contexts that differ from the transition context, nevertheless, we found these notions highly relevant). After providing further justification for the application of our model to transition in mathematics, we discuss its many implications in detail. By critically examining current practices, we enhance our understanding of the many issues involved in the transition. The core section 'Messages and implications of the model' is divided into subsections that were determined by the model (role of community, discontinuity of the transition process, shock of the new, role of time in transition, universality of transition, expectations and responsibilities, transition as a real event). Before making final conclusions, we examine certain aspects of remedial efforts.

Keywords: secondary–tertiary transition in mathematics; rite of passage; cognitive conflict; culture shock; bridge (remedial) courses

1. Introduction

There is no doubt that the secondary–tertiary transition in mathematics is a complex issue, involving a number of difficult situations (see, for instance, [1–10]). Perhaps the most notable feature of existing body of work on transition (in general, not just in mathematics, and not just secondary to tertiary transition) is the absence of a theoretical model. Our suggestion – to view the transition as a modern-day rite of passage – is an attempt at defining such a framework. With it, we argue, we are able

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to better judge the efforts aimed at dealing with issues of transition, as well as obtain a deeper understanding of the whole process[1].

There have been several attempts at critical evaluation of various efforts to mediate transition. For instance, an Australasian bridging mathematics network operated actively through the late 1980s to late 1990s/early 2000s. Although much of the work was published within their conference network and is hence not easy to source, some[2] of it is cited in [12]. Several studies presented at (or related to) Delta conferences[3] look critically at a variety of issues in transition (see, for instance, [1,13,14]). Nevertheless, further analysis and hard-core evidence (not just statistical) are needed. An example: without deeper understanding of the process, we cannot be sure that a bridging strategy that is genuinely successful at university A will prove itself successful when implemented at university B. We know very little about who attends such courses (self-selection, mandatory, etc.), and whether or not many of the attendees would be successful without them anyway. Furthermore, it seems that very weak students (who are, nevertheless, admitted to universities) do not benefit from bridging courses [10].

In [15] we began developing a model for the secondary – tertiary transition in mathematics, based on the anthropological notion of a rite of passage [16]. A rite of passage is a sequence of events that enables an individual to deal with, and overcome, a 'life crisis'. Present from earliest times and across all cultures and societies (losing their place somewhat in many modern societies), rites of passage accompany all major events in an individual's life (birth, betrothal, marriage, initiation, death, etc.). The stages present in a rite of passage mirror the general patterns of life (e.g. alive-dying-dead, child-teen-adult, etc.). Of particular interest is the middle ('dying', 'teen') phase, the so-called *liminal* phase (from *limen*, meaning threshold in Latin). In different cultures, it has been associated with powerful metaphors and symbols, such as death, being in the womb, invisibility, darkness, wilderness, or eclipse of the sun and moon [16].

In [15] we articulate the reasons why we believe that the educational transition from school to university mathematics should be viewed (and is) a rite of passage. The three phases of a rite of passage, applied to the educational transition, are:

- *separation* (from high school); this phase takes place while students are still in high school, and includes anticipation of forthcoming university life;
- *liminal phase*, also called *transition* (from high school to university); this phase includes the end of high school, the time between high school and university, and the start of first year at a university;
- *incorporation* (into university); this phase includes, roughly, the first year at a university.

In order to properly address all aspects of transition (such as a number of cognitive and pedagogical issues, for instance) we enrich our model with the notions of cognitive conflict and conceptual change [8,9,17–20], and culture shock [21]. Although defined and used in contexts that differ from the transition context, nevertheless, we found these notions highly relevant in this study.

Cognitive conflict and conceptual change refer to situations in which new knowledge (learned by a child, a student, or discovered by a research scientist) is incompatible with prior knowledge, and hence might affect understanding of the material. Based on a certain amount of information, an individual uses her/his own ideas and current understanding to create an initial explanatory framework.

The presuppositions that inhabit this framework, also known as naïve beliefs or alternative conceptions, will form her/his current beliefs about the material. Faced with having to absorb material that is in some way incompatible with prior knowledge (say, in a university math class), an individual will try to assimilate new information into their existing framework, thus creating a so-called synthetic model. As mixtures of beliefs and scientific facts, these synthetic models represent misconceptions about the subject. The robustness of synthetic models [22,23] is one of the major reasons why transition from high school to university mathematics turns out to be a difficult process for many students.

Finally, our use of the concept of culture shock (as developed for cross-cultural sojourns, i.e. longer stays in culture other than one's own, see [21]) was motivated by the similarity in emotional reactions to new situations[4]. In particular, in both travel and transition, a person undergoes a range of strong emotions (that influence her/his actions, behaviour, etc.), arranged in temporal sequence as [21, p. 80]

(a) the 'honeymoon' phase, initial reactions of euphoria
(b) the crisis, feelings of inadequacy and anxiety
(c) the recovery, including crisis resolution
(d) adjustment, reflecting enjoyment and functional competence.

Thus, concepts from several disciplines – psychology, anthropology, education, and cultural studies – intertwine mutually and contribute to our study and understanding of the parameters of the transition process in mathematics education. Having developed the model, we now provide further justification for its application to transition in mathematics. Incorporating concepts outlined in this section, we organize the discussion of messages and implications of our model according to the themes suggested by the model itself (role of community, discontinuities in the transition process, shock of the new, role of time in transition, universality and inclusiveness of transition, expectations and responsibilities of those in transition, and transition as a real event).

2. Rationale

Mathematics, mathematical thinking and applications of mathematics are integral parts of our daily life and our culture. However, 'different groups based on professional, social, cultural characteristics, race, ethnicity, gender, etc. have different views of mathematics and its role in the society' [5], and thus the transition becomes a multi-layered phenomenon, relevant not only to the future university students, but also to their families, and broader community, as well as society as a whole.

The report 'Tackling the Mathematics Problem' [24] prepared (in October 1995) by the Institute of Mathematics and its Applications, the London Mathematical Society and the Royal Statistical Society[5] states that

> 'There is unprecedented concern amongst mathematicians, scientists and engineers in higher education about the mathematical preparedness of new undergraduates. [. . .] The serious problems perceived by those in higher education are: (i) a serious lack of essential technical facility – the ability to undertake numerical and algebraic calculation with fluency and accuracy; (ii) a marked decline in analytical powers when faced with simple problems requiring more than one step; (iii) a changed perception of what mathematics is – in particular of the essential place within it of precision and proof.'

In the section 'A Brief Outline Of The Problem And The Required Response' of the same report[6], we read

> 'An equally serious concern of higher education is its observation of a qualitative change in the mathematical preparation of incoming undergraduates [. . .] This is no way restricted to those 'new undergraduates' who ten years ago would not have proceeded to higher education. The problem is more serious; it is not just the case that some students are less well prepared, but many 'high-attaining' students are seriously lacking in fundamental notions of the subject.'

Further evidence of significant changes in ability ranges of incoming students can be found in many references, in particular in [1].

The wide spectrum and sheer magnitude of the changes certainly qualify the secondary–tertiary transition as a rite of passage. In [19] the author states that

> 'Advanced mathematics, *by its very nature*, includes concepts which are subtly at variance with naïve experience. Such ideas require an immense personal reconstruction to build the cognitive apparatus to handle them effectively. It involves a struggle [. . .] and a direct confrontation with inevitable conflicts, which require resolution and reconstruction'.

The struggle 'with inevitable conflicts' is an integral part of the liminal phase of a rite of passage, and the 'resolution and reconstruction' characterize the incorporation phase. The pains and struggles of the liminal phase are illustrated in the following excerpt (note the 'pathway' metaphor)

> 'It is clear that the formal presentation of material to students in university mathematics courses – including mathematics majors, but even more for those who take mathematics as a service subject – involves conceptual obstacles that make the pathway very difficult for them to travel successfully' [19].

A 'reborn' individual (i.e. a person who has completed the rite of passage), attains knowledge, skills and attitudes that the new 'life' expects of her/him:

> 'And the changes in technology, that render routine tasks less needful of labour, suggest that the time for turning out students whose major achievement is in reproducing algorithms in appropriate circumstances is fast passing and such an approach needs to move to one which attempts to develop much more productive thinking' [19].

In many countries, high school curriculum used to be (to an extent) well defined, with precise descriptions of the material that needs to be taught, learning objectives clearly spelt out and with straightforward assessment methods. The outcomes were familiar to students, parents, teachers and everyone else. Moreover, the curriculum used to be more stable, i.e. did not change (in a significant way) every few years. Today the situation seems to be somewhat different. Numerous factors – frequent reforms in education, changes in mathematics curricula (in some cases described as[7] 'mile wide and inch deep'), complex (and therefore non-implementable) assessment methods, attempts at all-inclusiveness ('No Child Left Behind' Act[8], 'Raising the Floor' workshop[9]), a movement away from traditional secondary institutions (AP, IB courses and so-called 'cram courses') to mention a few – have created an educational landscape that is blurry, non-transparent and looks complicated (to say the least) to both insiders and outsiders.

There are many reasons why we should care about transition, and develop successful mechanisms to address the many problems associated with it. Besides the most obvious – improvement of our students' learning, increasing and/or maintaining their motivation to study mathematics and to consider a career in

a math-related field – we should make sure that the resources we have at our disposal (usually quite limited) are used most efficiently. For instance, is it better to pay an instructor to 'teach' a remedial algebra course, or should a university offer a summer preparation non-credit course, or should we just incorporate a review of algebra into the beginning of first year mathematics courses? If we decide on the first option, who should instruct: a high school teacher, university lecturer or a graduate student?

Moreover, we have a responsibility towards our students, as they strive to achieve their university goals. We should make sure that we do our best, i.e. that we offer resources that are the most effective for students' success. Students who fail in their studies will then know that it was not due to poorly designed transition and initial university offerings.

3. What is a successful transition?

The primary purpose of a rite of passage is to help the person pass from one well-defined, established and accepted position in life to another, which is equally well-defined, established and accepted. Thus, a success in transition can tentatively be defined as accomplishing most of: individual is comfortable in her/his new role as university student, she/he is able to achieve and work towards their goals, she/he shows good academic progress, she/he has support (both academic and otherwise), and can access it when needed, she/he enjoys mathematics courses, etc. However, students' individual goals and expectations of tertiary education are so broad that, by itself, would render futile any attempt at defining what success in transition really is.

What we certainly can claim is that the success depends, in great measure, on the robustness of certain parameters in secondary education (attitude, motivation, approach towards work, and, in particular, learning styles and cognitive models) that might need to be significantly modified at the tertiary level.

4. Messages and implications of the model

Based on the model that we have developed previously [15] and briefly outlined here, we now critically examine current practices and, in some cases, outline approaches and possible solutions to dealing with the difficulties of transition.

The transition from secondary to tertiary education involves both body and mind. One could argue that transition is a rite of passage in social, physical and psychological contexts, as well as in pedagogical and cognitive contexts. Acknowledging this fact, we prefer to use the term 'environment' rather than 'community' when talking about the totality of phenomena that surround an individual in educational transition.

4.1. *Community*

As we have just mentioned, by 'environment' we mean a totality of the contexts involved (social, psychological, cognitive, etc.). We will use the word 'community' when we are referring to a (predominantly) social context.

Talking about rituals that accompany rites of passage, Turner [25] notes that

> 'Among themselves, neophytes [participants in a rite of passage, op.a.] tend to develop an intense comradeship and egalitarianism. Secular distinctions of rank and status disappear or are homogenized'.

Here is a powerful message: students undergoing transition should be put together in a group (or several groups, if the total size is large), where there is no rule of selection. By virtue of experiencing the same fears and frustrations, and having to go through the same difficulties, the students will (or are expected to) start forming bonds, i.e. help each other. This process is very strongly dependent on cultural (societal) context. Later in this section we note that there are societies where the interaction between people is dominated by individualism[10]. Thus, in certain cultures, it might be necessary to stimulate and encourage individuals, through a variety of activities, to engage in a meaningful communication with their peers (which is the important initial step in community building).

Furthermore, those who have completed a rite of passage help prepare, in a multitude of ways, new cohorts that must undergo the same rite. Being (time-wise) close to completing it themselves, they are able to share their experiences and offer valuable insights to those who will be initiated. Learning from this, we conclude that older undergraduate students (peer helpers, mentors, big siblings) need to be a significant component of the community involved in the transition. In many cases, it is graduate students that are involved with first-year undergraduates. Although some are quite involved and interested in their students' success, for many reasons (age, cultural difference, lack of spare time, not living in the same space (dorms) etc.), they are more distant from first year students than, say, their second-year peers. These comments in [26, p. 15] echo our views:

> 'In providing assistance to first-year students there is a growing trend in the use of second- and third- year students in a mentoring role, supporting, but not replacing, experienced staff. Such student mentors go under a number of names [. . .] but the main idea is for them to pass on their experiences and help others with problems. Having gone through these problems themselves recently, they are in the position to understand and emphasize with struggling first-year students'.

We find further mention of the success of peer tutors/mentors in [10, pp. 94–95]. As participating in someone else's rite of passage contributes to the growth of that person, one's experience as a peer tutor/mentor becomes a significant component of her/his education process. (The best way to test whether we have learnt something is to teach it to someone else!) Thus, we recommend that the efforts of successful, motivating, engaged peer tutors be acknowledged in terms of academic credit, and not just financially.

Going a step further – Department of Mathematics and Statistics at McMaster University uses *qualified* undergraduate students as (formal) tutorial leaders in all first-year courses[11]. Besides holding formal tutorials, undergraduate teaching assistants are involved in all aspects of the course, both mandatory (office hours, marking tests, etc.), and optional (holding review sessions, organizing exam preparation workshops, etc.). First-time undergraduate teaching assistants are offered support in the form of a course (*Teaching Mathematics*) that they must take, and receive academic credit for[12]. For more information about the *Teaching Mathematics* course and details of work of undergraduate students as teachers, see [28,29]. One of measures of the success of this 'Teaching Mathematics' course is the

fact that several teaching assistants have decided to later pursue a teaching career (mostly in high school mathematics).

Recall that transition, classified as a rite of passage, includes (ideally) the time interval spanning the final year of secondary education and first-year university, including the temporal gap between the end of secondary schooling and the start of university classes. Knowing this, it is easy to identify the community for students in transition (our list does not suggest any form or ranking in terms of importance):

- Parents, siblings, spouses, partners, family, extended family, clan, church or social group
- Friends and peers
- Secondary school teachers, guidance counsellors
- University administration
- Instructors, peer tutors, teaching assistants
- Teaching and learning support centres at university
- Special-interest clubs and organizations, etc.

Clearly, one's community could easily include several dozen individuals. Community members involved in a rite of passage are familiar with the whole process, interested in its outcome, and fully aware of their roles and responsibilities. This fundamental principle, unfortunately, does not characterize the community related to the transition from secondary to tertiary education. Usually, this community consists of a number of individuals whose role in the whole process is not at all clear (and who might not be even willing to participate). As an example, in the next section, we explore the relationship between high school teachers and university instructors. Furthermore, there is almost no mention in the literature of the role of parents and family in the transition. Even those articles that talk about the community, and mention orientation programs, high school teachers, academic staff, teaching and learning centres, etc. do not mention students' family (this goes against basic principle of rite of passage, which requires active involvement of the *whole* community).

Rites of passage only happen within a clearly defined community. If we wish to make transition work, our first and most important goal is to build a functioning community, assign responsibilities to each member, and make sure that they all participate, and participate as expected. Community building is a difficult process, consuming time and energy. Some reasons for this lie in the existence of 'large cross-cultural differences in the pattern of reciprocal relationship between the individual and the group' ([21], p. 11) that are present in many cultures, and are especially dominant in almost all English-speaking countries. One way to measure the balance between the interests of the individuals and the group is to use the so-called individualism index [30]. Obtained by combining evaluations at three levels: personal (how individuals understand their identity), interpersonal or relational (how individuals define and function in social relationships, and whom they prefer to interact with), and societal or institutional (the nature of relationships that individuals build with the groups they belong to, and how they relate to authority), the individualism index turns out to be a number between 0 and 100, with 0 denoting the total absence of individual interests, and 100 the total absence of group interests[13]. Europe and North America occupy, not surprisingly, the top of this list[14] (US 91, Australia 90, UK 89, Canada 80, Netherlands 80, New Zealand 79, Italy 76, Belgium 75, Denmark 74, Sweden 71), with Asia and Latin America populating the

opposite end (bottom of the list: Guatemala 6, Ecuador 8, Panama 11, Venezuela 12, Columbia 13, Pakistan 14, Indonesia 14, Costa Rica 15, Peru 16, Taiwan 17, South Korea 18, Thailand 20, Singapore 20)[15].

The force of influence of community is echoed in

> '...neophytes [i.e. participants, op.a.] in many rites of passage have to submit to an authority that is nothing less than that of the total community. This community is the repository of the whole gamut of culture's values, norms, attitudes, sentiments, and relationships' ([25], p. 103).

Community is that entity that is fully trusted, that one can rely upon for help and support. Its existence is essential to the whole process.

4.2. *Discontinuities*

A rite of passage is a continuous process – once it has begun, it needs to be brought (through a sequence of well-defined steps) to an end; there are neither disruptions nor digressions in the process. In other words, at no time during a rite of passage the participants (or the community members) are left wondering – what do we do now? What's next? They know what the process is, or, if not, they are guided through it by a person in charge (consider, for instance the process of betrothal and marriage in any society or culture). Unfortunately, discontinuity is still one of major features of the transition process from high school to university mathematics. Let us focus on a few aspects of it.

Presently, the relationship between high school teachers and university instructors is far from satisfactory (almost every article on transition that we have examined stresses the importance of improvement of communication between the two groups). Although there are cases that contradict our statement [for instance, in some countries university math instructors are involved with writing and administration of high school final (or exit) exams], in general, very little useful information is exchanged between high school teachers and university instructors. When universities define high school prerequisites for their courses, they rarely consult with high school teachers. There is very little input (if any at all) that high school teachers provide in creation of syllabi for first-year university courses. Very few university instructors are aware of the material that is actually covered in high school classrooms. If, say, 'conic sections' is a topic listed in a high school curriculum, a university instructor might assume that students are familiar with concepts such as focus or asymptotes, when, in reality, students in high school might have only covered an ellipse (or just a circle).

In high schools in North America (and in some other countries), most courses are taught in a uniform way, and, as such, are (to an extent, of course) teacher-independent[16]. In university classrooms, students are exposed to different teaching styles that depend on a number of factors, such as the personality of the instructor and her/his beliefs, subject (calculus and linear algebra have a different 'flavour'), or level (first- *vs*. upper-year math classes)[17].

Certain practices used in the assessment and evaluation of students' work create further discontinuities. For instance, inflation of grades in high school[18] might affect the way a student perceives her/his abilities in mathematics. A high school student who believes, based on high school grades, that she/he is good at something (whereas in reality she/he is not), might lose quite a bit of confidence, self-respect and

motivation as they experience numerous difficulties in dealing with the material they thought they knew, or when they realize that their university marks are significantly lower than expected. Or if a high school exit exam is unreasonably difficult (as has happened in New Zealand)[19] they might choose not to attempt university mathematics.

Math tests in high school are, in many cases, graded with the aim of passing a student at almost any cost, rather than trying to provide a true learning experience, one from which a student would benefit later in university [5]. Similar practices (but perhaps motivated by different factors) occur in university classes as well. In [5] we find an interesting example: in marking test questions that involve the use of a theorem, some instructors neglect to reward checking assumptions that are of 'theoretical' nature. To be more clear: when using the Intermediate Value Theorem (in one-variable calculus) to show that the equation $f(x) = 0$ has a solution on an interval $[a, b]$, one needs to prove the continuity of f on the given interval, on top of checking that the values of f at the endpoints of the interval are of opposite signs. In some cases, students are given full credit if they checked the ('algebraic manipulation') assumption on the signs of f at the endpoints, and did not bother verifying the ('theoretical') assumption about continuity.

A number of university instructors make implicit assumptions about high school preparation of their students, in terms of content and skills. The most common assumption is that students are familiar with rudiments of formal logic (working with implications, understanding universal and existential quantifiers, etc.). It is interesting to note that many calculus books (in North America; perhaps elsewhere as well), that review basics such as rules of exponents or graphing linear functions, do not even mention rules of formal logic (that are used throughout, and in many situations, not even explicitly identified).

Further discontinuities arise in the sphere of the cognitive, and broader (pedagogy, philosophy), and are too numerous to be discussed or even enumerated here. For illustration, we mention a few here, and discuss some in more detail in the next section:

- nature of mathematics objects and reasoning about them; i.e. certain strategies that are efficient and successful in non-technical contexts are inadequate when applied to university mathematics [32]
- formal proof, and cognitive difficulties related to concepts such as limits, infinity, functions [8,9,20]
- development of strategies for reasoning, such as generalizing, property abstraction or working from definitions [32].

As another example of a discontinuity, we mention a deep issue that is certainly worth investigating – what is calculus actually? Calculus that is taught in high school (sometimes referred to as 'calculus for all') and the calculus taught at (most) universities are two different discourses. To understand transition better, we need to understand their differences.

4.3. *Shock of the new*

In almost all rites of passage (in particular, in initiation rites) pain – physical, but also psychological (isolation, fear, etc.) – is a significant component. In Maddern's

model [33] this would be classified as part of the 'challenge' stage, which can also include feelings of uncertainly, stress or confusion. Moreover,

> 'Liminality [actual transition, op.a.] implies that the high could not be high unless the low existed and he who is high must experience what it is like to below' [25, p. 97].

The person undergoing the passage needs to 'die', i.e. to abandon her/his old life, community, customs, responsibilities, etc. in order to be re-born into new community, that includes new customs, assumes new responsibilities and presents new challenges. The whole experience empowers the individual undergoing the rite, thus equipping her/him with necessary attitudes, skills, etc. that will enable them to function in the new community.

We note that in no rite of passage is an attempt made to bring the two communities (the one that the individual is abandoning, and the one she/he is being admitted into) closer. The symbolism of death in any rite of passage ceremony makes this perfectly clear. A successful transition program must keep this fact in mind. Rather than trying to 'ease the transition,' or 'make it smoother,' it needs to acknowledge that transition will be painful, difficult, and – perhaps most importantly – that it will take time. Students undergoing transition need to know and believe that all discomfort, pain, stress, even severe anxiety – in the end – will be proven worthwhile. Confusion and uncertainty are integral parts of everyone's learning process.

Next, we mention several cognitive and pedagogical 'shocks' that a student in transition will most likely be exposed to.

The passage from informal to formal language and reasoning is certainly one of most common cognitive problems that students experience when learning mathematics. Exploring the ways in which students work with definitions in mathematics, the authors state that

> '. . . certain reasoning strategies are inadequate when applied to university mathematics, although they might be efficient and sufficient in non-technical contexts and in the kind of reasoning with specific objects required by school mathematics' [32, p. 33].

University instructors intensify this problem by mixing colloquialisms and formal math language in their lectures. Switching from one to the other might be easy for a trained mathematician but is definitely an obstacle for students. Furthermore, many instructors are unaware of students' problems in working with formal objects:

> 'The student must learn to override these strategies [strategies mentioned in the previous quotation, op.a.] with the new approach of working from the dictated definitions, but since the role of mathematical definitions usually remains below the level of consciousness of working mathematicians, this is rarely communicated and might be far from transparent' [32, p. 33].

Working with theorems (understanding what a theorem says, understanding the implication (or equivalence) established by the statement, applying the theorem correctly, etc.) is equally challenging for students. Further major difficulties arise in dealing with proofs (see, for instance, [8,18,19]). In terms of learning strategies, perhaps the hardest obstacle is the passage from surface learning to deep learning strategies. Instructors need to spend considerable amount of time discussing proper, creative ways of learning mathematics, and include examples of situations where

surface learning approach fails. Bringing students' attention to misconceptions will further help in this process [34].

4.4. *Transition takes time*

The main reason why we decided to consider such a long incorporation phase in our transition scheme (ending roughly at the end of first year university) lies in the fact that any significant change requires time to be fully accepted and built into one's cognitive, social, psychological (and other) frameworks. Although not often, we do find the recognition of this fact in literature:

> 'Also note context and variety of forms $1-x(2-x)$ may be misread as $(1-x)(2-x)$, and $(2-x)(1-x)$ is significantly more difficult than $(x-2)(x+1)$ for weaker students, though to us they probably seem equivalent. So in this case students probably need lots of exercises (with answers), rather than teaching...' [26, p. 13].

Having to do *lots of exercises* takes time. One needs to recognize the fact that

> '...large amount of routine basic skills of mathematics is learnt only by repetitive practice; why should mathematics be different from any worthwhile activity? The motivation for such drudgery is that it will be useful in the end' [26, p. 16].

Furthermore, we need to keep in mind the time students will need to recall relevant background material. Losses due to absence of mathematics activity can be quite significant [35]. According to the study about the summer losses[20] [36] as reported on the Johns Hopkins National Center for Summer Learning [37] webpage[21]

> 'Most students lose about two months of grade level equivalency in mathematical computation over the summer months'.

An issue that requires more attention relates to university math success of students who followed an accelerated scheme in high school (skipping a grade, or taking a math course at some alternative institution[22]). There is data that shows that these students, although by many standards advanced in high school, do not do well in university, experiencing many problems that are related to transition [38]. By following the 'shortest path,' these students have no time to actually *do* mathematics, to think creatively about it, nor to internalize important ideas and concepts ('accelerated' in many cases means just going over the material faster, and does not involve necessary adjustments, say, in pedagogical approaches). Moreover, evidence shows that a very small number of 'accelerated' students opt for a major or minor in math – the subject that they were good at in high school. Quite often, they end up in computer science, economics, etc. with no desire to pursue math seriously [39]. Acceleration is rarely used in Japan, yet their secondary school pupils do very well comparative to other students [35].

Likewise, the purpose of accelerated recall sessions organized at universities ('cram' sessions, intense review sessions, etc.) needs to be examined. Although these sessions might work for some students (facilitating recall), they might not be very helpful (or not helpful at all) for students who did not learn the material in the first place. Furthermore, do they work in the long term, or is it a temporary fix at best?

Finally, to conclude this section – we have already asserted the value and importance of existence of community for individuals in transition. It is evident that

community building is a long-term project, and certainly cannot be accomplished in an accelerated rhythm.

4.5. *Transition is for all, and it affects the whole person*

There is no reason, and there is no way, to avoid going through a rite of passage. When the time comes (determined by a number of factors and/or defined by the community), an individual must undergo the passage. Likewise, secondary–tertiary transition is for all, even for those individuals who seem to be fairly comfortable in their new role as university students.

It is impossible to classify students early (say, during the first week of classes) according to the difficulties they might experience in transition to university. There is no evidence suggesting that for good high school students the transition is any easier, or differs in character, than for any other student group. There are students who excel academically (and otherwise) in high school, but whose transition to university environment is quite difficult, lengthy, and (in extreme cases) seriously affects their grades, motivation and confidence. A somewhat typical case[23] is a student who was well known and respected ('celebrity') for academic achievements in her/his small community, and who had hard time adjusting to being 'just one of' in a large metropolitan university. Success in transition is, to a large extent, independent of student's high school academic record:

> 'In themselves, entry qualifications are of limited use [. . .] Grades B to D at A level seem like a lottery, in that no knowledge of any particular topic can be relied upon' [40,41].

As well, university placement tests (or their equivalents), as presently administered at many institutions seem to be of limited use [42]. This comes as no surprise, if we recall that any rite of passage involves the totality of the individual undergoing the passage. No aspect of the personality (physical, emotional, social, etc.) is ignored or unaccounted for. In other words, an effective placement test (or other diagnostic tool) must inquire (besides obvious information on mathematics background knowledge and skills) about students' attitude towards mathematics and learning mathematics, about their motivation, intelligence, learning styles, their social environment, etc. – in order to capture the whole individual, and design appropriate activities for the transition.

4.6. *Expectations and responsibilities*

In [15] we state that

> 'A rite of passage assumes certain (again, well-defined) level of maturity of the individual involved, and demands that they take responsibility. We should not be shy in demanding that our students be responsible – in particular, and most important, in the context of their learning'.

Let us not forget that, ultimately, it is the student who has to negotiate the transition. It is the student who is responsible for her/his learning. Well-intended initiatives at helping students, if they go too far ('babying' students) will certainly disempower students as learners.

> 'A balance has to be struck between being too helpful, a short-term fix, and encouraging students to overcome their own deficiencies and problems' [26, p. 15].

As response to high failure and attrition rates, some universities have introduced compulsory one-on-one or small group tutorial activities. This seems like a good idea, but

> 'One might ask if this is really the way to breed independent lifelong learners endowed with mature self-discipline' [26, p. 15].

A rite of passage framework tells us that we need to clearly spell out the expectations that the course we teach places on the students. Equally importantly, we need to be explicit about our students' responsibilities with regards to their learning and the work in the courses we teach. We cannot assume that students are familiar, or know about it – as a matter of fact, evidence points to the contrary [26, pp. 5–6]. Early – before the concept of responsibility of a student as learner was discussed in class – students in a first-year science inquiry course at McMaster University were given a survey, i.e. an inventory of responsibilities related to their learning. Asked to select, for each item[24], whose responsibility they thought it was (instructor's, peer tutor's, or student's), very few students identified themselves as responsible. The situation changed dramatically after a serious discussion about the issue. Unfortunately, conveying the message once (or reiterating it) might not suffice. Even after the instructor[25] suggested that his calculus students should spend about 3 h at home for each hour of lecture (completing and studying notes, reading textbook, working on assignment questions, etc.), very few students have actually done so. In an informal poll, a great majority indicated that they study only about 2–3 h per week outside the classroom. In the National Survey of Student Engagement, as reported in [43], we read

> 'Only about 11 percent of full-time students say they spend more than 25 hours per week preparing for their classes – the amount of time that faculty members say is necessary to succeed in college. Forty-four percent spend 10 hours or less studying'.

As we said earlier, instructors can only go a certain distance. A student who ignores our efforts, advice and recommendations is no longer our responsibility.

In order to help transition in this respect, we suggest that high schools relax their structure a bit, to give more responsibility to students (for instance, by providing less information about tests, or making tests not completely resemble class notes; or, by asking students to learn about a topic from the textbook, with the follow-up lecture being a question-and-discussion session).

4.7. *Remedial efforts*

We have already asserted that addressing the transition in certain ways (say, by 'bringing the high school and university closer') might not be a good strategy. In this section, we briefly discuss – *only* in the context of our rites of passage model – remedial courses[26] as strategy to help students with the transition. We need to remember that, historically,

> 'A divergence of goals between secondary teachers and university lecturers has lead to the establishment of short-term "remedial" solutions in some universities' [44].

As we are all aware, these 'remedial' courses, originally driven by the stated difference in goals, have developed a life of their own. We need to understand how that 'life' has changed, and what role it plays today.[27]

> 'Goal of secondary teaching is to serve needs of society' [35].

Universities need to do this as well, but, besides that, their function is to provide future intellectual leaders in mathematics.

A very useful brief overview of remedial efforts (called 'bringing courses' there), including historic development and *raison d'etre*, can be found in [10]. What is absent in the article is what is, as a rule, absent in papers dealing with this issue – analysis of success and effectiveness in the long term (i.e. not measuring satisfaction and/or numeric success of students upon completion of a remedial course, but rather analysing their success in subsequent math courses that they take).

Rites of passage involve the whole person. In the same way, a remedial course (or similar effort) that has a single purpose (say, that aims at solely 'improving' students' algebraic skills) might not be effective at all. A successful course must aim to [10, p. 90]

- fill gaps in knowledge and skills, review essential facts and brush up on calculations
- prepare in terms of new approaches: (proofs, definitions, formal language, etc.)
- develop appropriate attitudes, language, learning styles, etc.

Furthermore, well-planned and organized remedial courses usually provide a good opportunity for its instructor to teach mathematics creatively, using a rich variety of strategies [10, p. 89].

Issues with remedial courses are too numerous to be discussed here, even at a superficial level. Nevertheless, we mention a few, that we use only to further illustrate the messages that we draw from our or rite of passage model.

In some cases, remedial courses have a stigma attached to them [26, p. 11]; [45, p. 202] which, of course, lowers students' self-confidence and motivation for further work in mathematics (such courses should be rethought or removed; no rite of passage ever leaves a participant in a state of shame or embarrassment, or lets her/him down – on the contrary, it empowers the participants in multiple ways). Referring students to use computers to do remedial work is rarely beneficial [26, p. 14]. Even with our best intentions we still lack conclusive research on ways in which modern technology possibly enhances one's learning; i.e. the environment, in the sense of rite of passage, is unclear and not well understood – which no rite of passage allows to happen. The fact that, in many cases, remedial courses do not help weak students [10, p. 92] indicates a failure in design of such courses; a remedial course (like a rite of passage) needs to be for all, not just for some or for a majority.

We conclude this section by presenting one case – data on students' subsequent completion of math courses after taking a remedial course. (Tracking of students through foundation and first-year courses has also been done at University of Auckland[28]. We are interested in learning if similar data exists and/or has been presented elsewhere). The course in mathematics and statistics that we comment on here, offered at Victoria University[29] – although remedial in nature – is often referred to as the Foundation course. Passing grades in the course are C, C+, B−, B, B+, A−, A, A+, in ascending order. In general, the C's are bare passes and correspond to scores of between 50 and 59%, whereas the three A grades cover scores from 75 to 100%. Thus, grades of B− or higher attest to a solid grasp of the material.

In 2005, of the 133 students registered for the Foundation course, 40 students failed, while the remaining received passing grades, in the full range between C and A+. Disregarding the C grades and focusing on students with grades of B− or higher, i.e. on those students who might reasonably believe that they were well

prepared for university courses in mathematics, we notice that the success for this cohort in the subsequent year is not satisfactory (see Appendix for details). The students who gained passes in the Foundation course at a level well above a bare pass had a pass rate the following year of 50% or less[30]. Moreover, high grades in the foundation course did not correspond to high grades the following year with A− and B− students doing better than A or B+ students respectively.

We are aware that this data represents one cohort of students only, and might not be representative of large samples. However, it raises serious questions about the efficacy of such courses.

4.8. *Transition is a real event*

All rites of passage deal with *real* events in one's life, and they reflect the *reality* of the context (social, physical, geographical, etc.) in which they take place. Unfortunately, there are important aspects in which the context for transition in mathematics does not reflect reality. Let us explore some of these aspects.

According to an international survey, one of the major reasons for the gap between secondary and tertiary education has been identified as 'Too optimistic assumptions and expectations of university lecturers' [5]. Student populations have been changing dramatically, at an ever-increasing pace over the last few decades. Perhaps the most significant (and universal) changes are in the decreasing levels of preparedness of students for the tertiary education, and in the reduction of the amount of time students are able/willing to dedicate to their studies. Trying to understand and control the forces that cause these changes (frequent, major curricular changes, ill-conceived and/or politically-motivated reforms of elementary and secondary education, large number of students holding part-time or full-time jobs that they need to pay for increasing costs of education, etc.) might not be as productive at this moment as trying to understand how they shape the students, that today, in 2008, sit in our classrooms. Furthermore,

> 'Our current student group, as well as having a great variety of qualifications and ability, also have variety in social background, age, work experience and ambitions or intentions' [26, p. 6].

This is the reality that we need to work within, and incorporate into our transition (and general educational) efforts. The magnitude of changes are well portrayed in the following excerpts from 'Digital Natives' [46]

> 'Today's students have not just changed *incrementally* from those of the past, nor simply changed their slang, clothes, body adornments, or styles, as has happened between generations previously. A really big *discontinuity* has taken place. One might even call it a "singularity" – an event which changes things so fundamentally that there is absolutely no going back. This so-called "singularity" is the arrival and rapid dissemination of digital technology in the last decades of the 20th century'.

Students that are coming to universities these days have grown up with digital technology:

> 'They have spent their entire lives surrounded by and using computers, videogames, digital music players, video cams, cell phones, and all the other toys and tools of the digital age. Today's average college grads have spent less than 5000 h of their lives reading, but over 10,000 h playing video games (not to mention 20, 000 h watching TV). Computer games, email, the Internet, cell phones and instant messaging are integral parts of their lives' [46].

It is reasonable to conclude that, as a consequence, students' cognitive processes have been affected in a fundamental way.

To account for changes (not just) in student population, the transition process itself will require constant readjustments (as all rites of passages do over time; for instance, compare the marriage ceremony in almost any society/group/culture today to that of 50 years ago). We can, most certainly, learn from successful strategies used to address the transition twenty years ago (bridging courses, orientation sessions, etc.), but we cannot implement those without any changes today, and expect to be equally successful.

As part of their incorporation process, students need to understand that university life is not in any way ideal (How many times have we heard a student say 'This is not fair'?). Not all exams and tests are marked in *exactly* the same way. Not every section of the same course has tests of *exact* same level of difficulty, etc. Reflecting the reality of life, in university, students will find themselves 'oscillating' between equality and inequality, fairness and unfairness, academic honesty and dishonesty, homogeneity and differentiation, success and failure. In spite of best efforts, bias, forms of racism, ageism, homophobia, or tension between various groups (formed on basis of ethnicity, culture, religion, or belonging/not belonging to a program viewed as 'elite') will always be present on university campuses.

The following passage gives a fairly accurate portrait of a large number of university instructors:

> 'University lecturers were not typical students even at the time when they were students, and, in many cases, that was 10, 20, or even 40 years ago. We liked our subject, worked adequately if not hard, and also perhaps had a style of learning that was atypical. Understanding what we were doing, finding out for ourselves, checking our results [. . .] were likely to be our characteristics' [26, p. 5].

Assuming that our students (or great majority of them) are 'like us' is one of the most common misconceptions that many of us hold. A typical instructor, when preparing a course, thinks of it as living in 'vacuum,' as a box disconnected from other boxes. Assuming that students know all material that was covered in prerequisite courses, not inquiring about the backgrounds of students and their reasons for taking the course, not planning to use any kind of diagnostic tool, or student feedback mechanism, an instructor will create a course that, in many ways, is *unreal*; it might be frustrating to teach, and students might perceive it as too difficult, or not interesting, or not relevant. Furthermore, certain forms of instructor behaviour can only be deemed as irrational. For instance, even when they are fully aware of their students' lack of background knowledge – 'My students don't know what linear independence means', 'They have no idea what a vector space is', etc. – instructors will, nevertheless, proceed with lecturing as if the students knew how to prove linear independence or are able to recognize a vector space. Another aspect of ignorance is re-teaching the material that students are familiar with. In [47] Young-Loveridge states that, in New Zealand classrooms

> '. . . while teachers in new entrant classes have a very clear idea of the number skills and ideas children have when they start school, they don't in fact then teach new skills and 80% of children were retaught skills they already had'.

Further evidence is found in [48]. Sherley [49] is finding evidence consistent with Young-Loveridge, and Averill [50] finds a similar pattern in the shift from

elementary to high school. Knowledge of prior learning should enable university instructors to begin at a realistic level.

5. Conclusion

Using our rite of passage model, we have been able to comment on and criticize a number of strategies aimed at addressing a whole variety of issues in the transition from secondary to tertiary mathematics. We believe that this approach can be used to assess other transitions, both in mathematics (transition into higher years in university, transition between grades in high school, etc.) and elsewhere. Our future work will focus on the refinement of several ideas presented here.

Perhaps the most important message that our model suggests is that the transition cannot happen in the absence of a proper environment, and it is our duty to identify who and what constitutes that environment and to find ways of building it. Its continuous presence will guarantee that an individual will be able to successfully complete the passage into tertiary mathematics. Furthermore, our model suggests that we need to think about replacing the temporal gap between secondary and tertiary education with a set of meaningful activities.

Can universities afford[31] not to offer remedial courses? Perhaps, by not teaching remedial mathematics, universities might be able to force high schools to better prepare their students for university. If we continue the present trend of offering remedial courses (that, in some cases, earn university credit) at lower and lower levels, what are we going to teach in university ten, twenty years? Alternatively, would it not be better not to offer such courses but to alter our present first-year offerings so that we can best build on the students' foundations?

There remain some difficult questions: Why is the gap between secondary and tertiary mathematics so wide? Why is it that we need to abandon so many high schools practices, attitudes, etc., rather than build upon those foundations when we teach university courses? These questions require answers before a truly successful transition strategy can be created.

We thank the reviewers for their insightful and detailed comments and suggestions that helped us improve this article.

Notes

1. This is not the first time an anthropological model has been used in a mathematical context. See [11] for such an approach to number and time.
2. Such as contributions by Boland, Laird, and Taylor.
3. Biennial Southern Hemisphere conferences in mathematics educations, whose primary focus is tertiary education.
4. One could argue that there is a large difference in a successful (it is not at all clear what 'successful' really means) outcome between a cross-cultural sojourn and transition: the adoption, and creation of the balance between one's own and 'new' cultures for the former, and expectations that students reject their 'high school' culture (say, in terms of various parameters of their cognitive environment), and accept the new, 'university' culture for the latter.
5. The quote is from the *Summary* section of the report.
6. Third paragraph.
7. Common phrase used to critique US mathematics curricula (K-12) that tend to include too many topics; sometimes blamed for students' poor achievement in mathematics.
8. Introduced in January 2001. Available at: http://www.ed.gov/nclb/landing.jhtml.

9. Raising the floor: Progress and setbacks in the struggle for quality mathematics education for all, workshop, May 7–10, 2006. Available at: http://www.msri.org/calendar/workshops/WorkshopInfo/388/show_workshop.
10. The individualism in certain cultures is fed by (or feeds) competitiveness - strong negative force, working against community forming. There are cases where students refuse to lend their class notes to a peer who missed the class; or (anecdotal evidence from McMaster University) a student would knowingly give an assignment containing wrong solutions to her/his peer.
11. McMaster University is by no means unique in this way. For instance, University of Auckland in New Zealand has been running a similar initiative for many years; see [27].
12. They also get paid for their job as teaching assistants.
13. Thus, on average, in a country with individualism index of 30 people believe and rely much more on group/collective values than people living in a country with the index of 70.
14. One could comment that this seems to be counter to contemporary practices in teaching mathematics (predominantly in the Western world) that foster group work and peer communication. However, anecdotal evidence (such as students opting out of group work if given the opportunity, as witnessed by a co-author of this article, M. Lovric in his problem-solving course at McMaster University) as well as research indicate that group work is not readily embraced by students (i.e. they need to be stimulated to engage in group work), nor are its benefits evident and/or present. For instance, examining practices in teaching in high school in Ontario, Canada, the author demonstrates that 'Students perceived as low achievers never become legitimate participants in the peer discourse, despite their efforts.' [31, p. 470].
15. China was not surveyed.
16. Of course, students' experiences in high school (as well as in university) mathematics are strongly teacher-dependent. Here we refer to the rigidity of a high school curriculum, as compared to a more flexible coverage of the material in a university course.
17. Even more than that: material covered (i.e. choice of topics, as well as decisions about what theorems will be proved, or what applications will be presented) in a given course might depend on the course instructor.
18. We will not go into discussing reasons for it here. Probably the best evidence of this common practice is the fact that high school students in Ontario (and perhaps elsewhere) are told (by guidance counselors, older peers, or others), to expect that in university their grades will 'drop by about 30%'.
19. The year in which this effect was noticed was 2001, with consequent dramatic lowering of first year calculus enrolments of up to 20%. At the time, concerns to the Qualifications Authority were expressed by both Victoria University and Auckland University.
20. Related to students who live on the Northern hemisphere, i.e. whose school vacations take place between June and August.
21. About the Center > Know the Facts.
22. 'Going to school after school'.
23. Anecdotal evidence from McMaster University.
24. Sample of items from the inventory: assess prior knowledge, set goals, identify resources, assess progress, develop critical attitude, construct knowledge, scheduling time, etc.
25. Miroslav Lovric, co-author of this article.
26. By 'remedial' we mean courses that prepare students 'for the academic concepts, skills and attitudes of their degree programmes as opposed to orientation programmes that prepare students for university life' [10, p. 89]. Such courses are also called bridge or bridging courses, foundation courses, access courses, pre-calculus courses, etc. This is sometimes confusing, as some other types of courses, which are not remedial in nature (e.g. courses that introduce students to formal mathematical reasoning, or proof), are occasionally called foundation or bridge courses.
27. In this section, we discuss several important contributions in that direction. Our discussion does not aim to be comprehensive.
28. Auckland, New Zealand. The examination marks system used in Department of Mathematics at the University of Auckland allows for comparisons of results from one course to another, as well as looking at students' performance as a group.
29. Wellington, New Zealand.

30. Students who took the Foundation course (along with many other first year university students) had an interview with a mathematics department member in order to identify the most suitable course for them from the list of available first-year mathematics courses. The issue was that the Foundation course grades were so unreliable as predictors that it became very difficult to give sound advice. On the contrary, the grades students received in secondary school mathematics turned out to be a fairly good indicator of what first-year university math course is most appropriate for them.
31. Large financial gains aside, for the moment.
32. Students need to complete 13 years of elementary and high school education before coming to university.
33. As mentioned earlier in the text, the course – although referred to as 'foundation' course – is actually remedial in nature.
34. Paper is a course.

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Appendix

Explanation for entries in Table A1: pass grades, in descending order are: A+, A, A−, B+, B, B−, C+, C; fail grades: D, E and Q (failed to complete). Grade of W is given to a student who withdrew the paper (course). MATH103 and MATH 104 are first year courses for students with year 12 mathematics as their highest level, and also for those who showed poor performance in year 13 mathematics[32]. MATH113 and MATH 114 are usual first year math courses for students with year 13 mathematics. STATS 193 is a service course in statistics for natural and social science majors. The symbol n in the second row denotes the number of students attaining the particular grade in the foundation course in 2005.

Table A1. Grades in math courses attained in 2006 by Victoria University students who completed the foundation course in 2005.

		Grade (number of students) in the foundation[33] course in maths and stats, 2005			
		A ($n=4$)	A−($n=8$)	B+ ($n=17$)	B− ($n=13$)
Grades in Math Papers[34] in 2006	No further math papers taken	1	3	5	5
	MATH 103	B−, Q		D, E	B, D
	MATH 104	C, C	C	C+	B−, E
	MATH 113	W	E		W
	MATH 114	E, E	E, E, E	D, Q	C, W
	STAT193	E	B−, C	B, B−, B−, E, Q	B+, D, D, E
	Commercial math		B+	B+, E, E	B
Summary	Papers attempted	8	8	13	12
	Passes	3	4	5	5
	Fails (W grades not included)	4	4	6	5
	Pass rate of attempted	37.5%	50%	38.5%	42%